

JAVA PROGRAMMING

SCENARIO BASED QUESTIONS

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Your Code

```
1 import java.util.*;
2 class StudentMarks{
3     public static void main(String[]args)
4     {
5         int[]marks={45,67,89,32,76,50,91,84,60,73};
6         int highest=marks[0];
7         int sum=0;
8         for(int i=0;i<marks.length;i++){
9             sum+=marks[i];
10            if(marks[i]>highest){
11                highest=marks[i];
12            }
13        }
14        double average=(double)sum/marks.length;
15        int count=0;
16        for(int i=0;i<marks.length;i++){
17            if(marks[i]>average){
18                count++;
19            }
20        }
21        System.out.println("highest="+highest);
22        System.out.println("average="+average);
23        System.out.println("student above average="+average);
24
25    }
26 }
```

Input

stdin input

Output

highest=91
average=66.7
student above average=66.7

Visible Test Cases

Test Case 1

Test Case 2

Test Case 3

Your Code

```
1 import java.util.*;
2 class Attendance{
3     public static void main(String[]args)
4     {
5         String attendance="PPAAPPAPPP";
6         int present=0;
7         int absent=0;
8         for(int i=0;i<attendance.length();i++){
9             if(attendance.charAt(i)=='P'){
10                 present++;
11             }
12             else if(attendance.charAt(i)=='A'){
13                 absent++;
14             }
15         }
16         double percentage=(present*100.0)/attendance.length();
17         System.out.println("present="+present);
18         System.out.println("absent="+absent);
19         System.out.println("attendance="+percentage+"%");
20         if(percentage>=75){
21             System.out.println("eligible");
22         }else{
23             System.out.println("not eligible");
24         }
25     }
26 }
```

Input

stdin input

Output

```
present=7
absent=3
attendance=70.0%
not eligible
```

Visible Test Cases

Test Case 1

Test Case 2

Test Case 3

```
1 import java.util.*;
2 class ExamResult{
3     public static void main(String[]args)
4     {
5         int[]marks={45,67,76,50};
6         int pass=0;
7         int fail=0;
8         for(int i=0;i<marks.length;i++){
9             if(marks[i]>=50){
10                 pass++;
11             }else{
12                 fail++;
13             }
14             if(marks[i]>=90){
15                 System.out.println(marks[i]+"grade A");
16             }
17             else if(marks[i]>=75){
18                 System.out.println(marks[i]+"grade B");
19             }
20             else if(marks[i]>=50){
21                 System.out.println(marks[i]+"grade C");
22             }
23             else{
24                 System.out.println(marks[i]+"fail");
25             }
26         }
27         System.out.println("pass="+pass);
28         System.out.println("fail="+fail);
29     }
30 }
```

stdin input

Output

```
45fail
67grade C
76grade B
50grade C
pass=3
fail=1
```

Visible Test Cases

Test Case 1

Test Case 2

Test Case 3

ram, Rita, Ram

Test Cases:

1. {A,B,C}
2. {Tom,Tom,Tom}
3. {X,Y,X,Z,Y}
4. {A,A,B,B,C,C}
5. {Raj,Raj,Raj,Raj}

Your Code

```
1 import java.util.*;
2 class Main{
3     public static void main(String[] args)
4     {
5         String[]a={"ram","sita","ram","john","sita"};
6         HashMap<String,Integer>m=new HashMap<>();
7         for(String s:a)
8             m.put(s,m.getOrDefault(s,0)+1);
9         for(String s:m.keySet())
10             if(m.get(s)>1)
11                 System.out.print(s+"="+m.get(s)+" ");
12     }
13 }
```

Input

stdin input

Output

sita=2 ram=2

Visible Test Cases

5. {100,-100,50,-25}

Your Code

```
1 java.util.*;
2 main{
3     public static void main(String[] args)
4     {
5         int[]a={500,-200,1000,-300,700};
6         int bal=0,dep=0,wit=0;
7         for(int i:a){
8             bal+=i;
9             if(i>0)dep++;
10             else if(i<0)wit++;
11         }
12         System.out.println("balance="+bal+"deposits="+dep+"withdrawals="+wit);
13     }
14 }
```

Input

stdin input

Output

balance=1700deposits=3withdrawals=2

Visible Test Cases

Test Case 1

Test Case 2

Test Case 3